



DANISH KASHAEV

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PERSONAL INFORMATION

Name: Danish Kashaev

Nationality: Swiss

Date of birth: 03.04.1997

EDUCATION

PhD Student <i>Networks and Optimization</i>	2021 – 2026
Centrum Wiskunde & Informatica University of Amsterdam	Amsterdam, Netherlands
Master of Science <i>Mathematics (5.8/6)</i>	2018 – 2021
ETH Zurich	Zurich, Switzerland
Exchange Student	2017 – 2018
University of California	Santa Barbara, USA
Bachelor of Science <i>Mathematics (5.6/6)</i>	2015 – 2018
University of Geneva	Geneva, Switzerland
Gymnasial Matura (Bilingual English)	2011 – 2015
Collège et Ecole de Commerce André-Chavanne	Geneva, Switzerland

WORK EXPERIENCE

Teaching Assistant: Mathematical Optimization	2020
ETH Zurich, Institute for Operations Research	Zurich, Switzerland
- Led a two-hour weekly exercise class for the 11 ECTS course Mathematical Optimization - Graded homework assignments - Grading of the exam	
Algorithm Developer (Part-Time)	2021
Aspaara Algorithmic Solutions	Zurich, Switzerland
- Developed algorithms for applied optimization problems: assignment, scheduling, mixed integer programming - Implementation of the algorithms in Python	
Teaching Assistant: Algorithmic Game Theory	2023
University of Amsterdam	Amsterdam, Netherlands
- Led a two-hour weekly exercise class for the course Algorithmic Game Theory - Graded homework assignments - Grading of the exam	

THESES

Master's Thesis

- ETH Zurich, Institute for Operations Research
- Title: *An Optimal Monotone Contention Resolution Scheme for Uniform and Partition Matroids*

PhD Thesis

- University of Amsterdam, Institute for Logic, Language and Computation
- Title: *Approximation via Duality in Offline, Online and Strategic Settings*

A Simple Optimal Contention Resolution Scheme for Uniform Matroids

- Co-author: Richard Santiago
- Journal version in *Theoretical Computer Science* 2023
- ArXiv version: <https://arxiv.org/abs/2105.11992>

A Nearly Optimal Randomized Algorithm for Explorable Heap Selection

- Co-authors: Sander Borst, Daniel Dadush, Sophie Huiberts
- Journal version in *Mathematical Programming* 2024
- Conference version in *IPCO* 2023
- ArXiv version: <https://arxiv.org/abs/2210.05982>

Round and Bipartize for Vertex Cover Approximation

- Co-author: Guido Schäfer
- Conference version in *APPROX* 2023
- ArXiv version: <https://arxiv.org/abs/2211.01699>

Online Matching on 3-Uniform Hypergraphs

- Co-authors: Sander Borst, Zhuan Khye Koh
- Journal version in *Mathematical Programming* 2026
- Conference version in *IPCO* 2025
- ArXiv version: <https://arxiv.org/abs/2402.13227>

Selfish, Local and Online Scheduling via Vector Fitting

- Conference version in *SODA* 2026
- ArXiv version: <https://arxiv.org/abs/2505.10082>

Improved Online Load Balancing in the Two-Norm

- Co-author: Sander Borst
- ArXiv version: <https://arxiv.org/abs/2511.03345>

ATTENDED WORKSHOPS AND CONFERENCES

Workshop on Algorithms with Predictions

2022, Ecole Polytechnique Fédérale de Lausanne

International Conference on Integer Programming and Combinatorial Optimization

2022, TU Eindhoven

LNMB Conference on Mathematics of Operations Research

2023, Soesteberg

International Conference on Approximation Algorithms for Combinatorial Optimization Problems

2023, Georgia Institute of Technology

Aussois Workshop on Combinatorial Optimization

2024, Aussois

Fulkerson 100 Workshop

2024, University of Waterloo

International Symposium on Mathematical Programming

2024, Montréal

Cargèse Workshop on Combinatorial Optimization

2024, Cargèse

Summer School on Synergies of Combinatorics and Theoretical Computer Science

2024, Ecole Polytechnique Fédérale de Lausanne

Highlights of Algorithms

2025, ETH Zurich

International Conference on Integer Programming and Combinatorial Optimization

2025, Johns Hopkins University

Satellite Workshop: Learning Augmented Algorithms

2025, CWI Amsterdam

ACM-SIAM Symposium on Discrete Algorithms (SODA)

2026, Vancouver

SKILLS

Languages

French (native), Tatar (native), English (fluent, TOEFL: 113/120), Russian (fluent), Dutch (B2-C1)

Programming

Python, C++, R, MATLAB, Mathematica

Document Creation

LaTeX

EXTRACURRICULAR ACTIVITIES

Tennis

Highest Swiss ranking: N4

Music

Geneva Music Conservatory: *Piano* (14 years), *Solfège* (4 years), *Music theory* (3 years), *Piano improvisation* (3 years)